

Water Cycle

The Hydrological Cycle

Define the hydrological cycle and explain how it operates at both the global and catchment scales. In your answer, describe three key processes and how they interact across these scales.

Water Balance in a Semi-Arid Basin

Consider a large basin in a semi-arid climate. Using the water balance equation

$$P = Q + E + \Delta S$$

discuss how each term is likely to behave in this setting. Explain what challenges this poses for hydrologists in measuring or estimating water availability.

Global water distribution

Let's go back to a quote from the lecture:

“The total quantity of fresh water on Earth could satisfy all human needs if it were evenly distributed and accessible.” — Stumm (1986)

Critically evaluate this statement using information from the lecture and Chapter 1. Include examples of spatial inequality in water resources and discuss how hydrology as a science and applied field contributes to addressing such challenges.

Delineating a Watershed Boundary

Refer to the **topographic map provided** below. The contour lines represent elevation. A stream is visible in the valley. Your task is to manually delineate the **watershed boundary** for a given outlet point along the stream using standard hydrological reasoning.

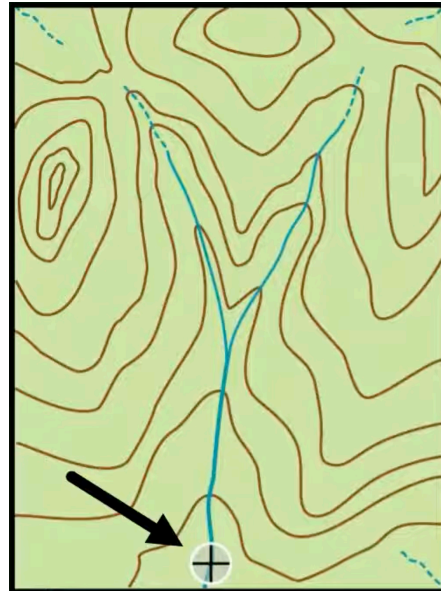


Figure 1: A hypothetical topographic map of a catchment

1. Briefly list the **five manual delineation steps** for defining a watershed boundary.
2. On the map, **describe** (in words) how you would apply each step. You may refer to locations by compass direction (e.g., “north ridge”) or elevation.
3. Explain why understanding **topographic divides** and **contour orientation** is critical in watershed boundary definition.
4. (Optional extension) What is a limitation of this method compared to digital (GIS-based) watershed delineation?